# A Study of Efficient User Representation Computation and Maintenance in Sequential Recommender Systems

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**Abstract**

Sequential recommender systems use interaction histories to represent user interests. Recomputing these representations after every interaction can be expensive, while reusing old representations may reduce recommendation quality. This study will compare strategies for refreshing user representations in music recommendation on Yambda. The model will combine a cached long-history representation with a small encoder of recent activity. Experiments will assess simple update schedules, a lightweight adaptive controller, and reuse of intermediate transformer states. The goal is to reduce computation while preserving recommendation quality.

## 1 Introduction

Sequential recommender systems predict future actions from past interactions. Transformer models such as SASRec encode an ordered item history into a user representation [1]. Larger models can capture longer histories, but require more computation [2].

A practical question is how often these representations should be updated. Frequent updates include the latest activity but repeat expensive computation. Less frequent updates save resources but may miss changes in user interests. A small encoder of recent interactions can provide current information while an older long-history representation supplies broader context.

This study will investigate three questions: how much representation freshness improves recommendation quality, whether adaptive updates outperform simple schedules, and when reusing intermediate transformer states is more efficient than recomputing the representation. The working hypothesis is that the value of an update varies across users and prediction opportunities.

## 2 Novelty and Contributions

The expected contributions are:

- **A comparison of update strategies** on public music data, measuring both recommendation quality and resource consumption.
- **An analysis of representation freshness**, showing when old long-history representations remain useful alongside recent activity.
- **A lightweight adaptive update method** that estimates whether an update is useful from recent activity and cache age, if the initial experiments show sufficient benefit from freshness.

Caching and long-history/recent-history architectures are established ideas. The potential methodological contribution is predicting the benefit of an update and using it to allocate computation. Its effectiveness and novelty will be assessed through experiments and literature review.

## 3 State-of-the-Art

- **SASRec** [1] uses self-attention for sequential recommendation and will be the starting point for the implementation.
- **ARGUS** [2] studies large recommendation transformers and supports offline computation of user representations, motivating the need for efficient maintenance.
- **Long-History User Transformers** [3] combine a long-history representation computed asynchronously with an encoder of recent events. This provides the architectural basis for the study.
- **TransX** [4] reuses behavior encodings and combines incremental updates with periodic processing. It motivates activity-based scheduling as a baseline.
- **HSTU/M-FALCON** [5] reuse shared history computations, including key–value (KV) states, providing an alternative to caching only the final user vector.
- **Yambda** [6] provides public music interaction data and a recommendation benchmark suitable for the planned experiments.

## 4 Goal of the Thesis

The goal is to **identify methods for maintaining user representations that reduce computation while preserving next-item recommendation quality**.

The experiments will use Yambda-50M and a SASRec-style model with two branches: a long-history encoder whose output is cached, and a smaller encoder that processes recent interactions before every prediction. Their outputs will be combined to score candidate tracks.

The main experiments will:

- Measure how recommendation quality changes with cache age and compare recent-only, long-history-only, and combined models.
- Compare updating at every prediction, keeping the initial cache, and schedules based on time, event count, or user activity. If freshness matters, evaluate a small controller that predicts the benefit of updating.
- Test KV-state reuse, checking that it produces equivalent outputs under matching history conditions and measuring update time and memory. Account for rebuilds when changes to the history invalidate retained states.

Evaluation will replay interactions chronologically, using only past events at each prediction. Update strategies will share a fixed recommender; controller training and policy tuning will use periods preceding the final test. Quality will be measured with NDCG@10 and Recall@100 on positive listening targets. Efficiency will be measured through processing time and memory, including recent-history encoding, candidate scoring, and cache handling. The results will describe offline predictive performance.

# 5 Work Plan and Schedule

The work is planned for October 2026–March 2027.

- **October 2026:** Literature review and data preparation.
- **November 2026:** Model and evaluation pipeline implementation.
- **December 2026:** Initial experiments on representation freshness.
- **January 2027:** Update strategies and KV-state reuse experiments.
- **February 2027:** Comparative evaluation and analysis of results.
- **March 2027:** Thesis writing, revisions, and finalization.

# 6 Conclusion

The study will establish when user representations need to be refreshed and which maintenance strategies offer a useful balance between quality and cost. The expected outcome is an experimentally supported approach to efficient user-state maintenance on public music data. If adaptive updates provide little benefit, the study will identify the simplest adequate strategy.

# References


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